

LabX & CDS — Complete Cheat Sheet

Cognizant Training Reference | LabX License, User Management, Instruments, Workspaces, CDS Basics, OpenLab CDS & ECM

1. LabX — INTRODUCTION & PURPOSE

What is LabX?

LabX (by METTLER TOLEDO) is a laboratory data management and instrument control software enabling centralized control, monitoring, and documentation of results from balances, titrators, pH meters, density meters, refractometers, melting point systems, and Karl Fischer titrators.

- Automated data collection, standardized procedures, secure handling of results.
- Supports electronic records and audit trails — ideal for GxP, GLP, 21 CFR Part 11 compliance.

Purpose

- Centralized instrument control and method management
- Automatic result collection and reporting
- Seamless integration with LIMS, ERP, and other lab systems
- User management with roles, permissions, and audit trails
- Enhanced data quality, integrity, compliance, and productivity

2. SYSTEM SPECIFICATIONS & ARCHITECTURE

System Specs

- Client/Server system — every attached computer and instrument is treated as a client.
- Express Edition: up to 3 instruments on one local computer (Standalone).
- Server Edition: distributed or localized, up to 30 instruments per server.
- Clients: Windows executables — NOT web-based. Servers: hosted in Windows services.
- Database: MS SQL Server or Oracle.

Architecture Modes

- Standalone: All components (DB, app, service) on one computer.
- Distributed (Preferred): Components spread across separate computers.
- Business Server connects to at least 3 Instrument Servers; each connects 7–15 instruments.
- Instruments connect via Ethernet — no physical lab PC needed per instrument.
- LabX v11: up to 30 instruments (or 60 with Server Full Edition) per MS SQL DB instance.

Key Components

- Client-Server: UI on workstations; server manages data, processing, instrument comm.
- Communication: USB, Ethernet, Serial (industry-standard protocols).
- Security: User auth, data encryption, audit trails.
- Integration: API-based with LIMS and other systems.
- Scalability: Supports multiple users/instruments; future-ready.

3. LICENSE MANAGEMENT & ACTIVATION

License Procurement — Step by Step

Step 1: Choose Starter Pack — Express (local, ≤3 instruments) or Server (30/60 instruments, multiple computers). With Express-Server license you can Sidegrade Express to Server.

Step 2: Configure Options — User Management, Product Database, Auto Import/Export, Statistics/Trending, Regulation (21 CFR/EU Annex 11, Server only), Report Designer, System Integration (API/LIMS/ELN/ERP), STAReX Integration.

Step 3: Choose Instruments — LabX licensed per connected instrument. Supported: Analytical/Precision Balances, Titrators, Karl Fischer, pH Meters, UV/Vis, Refractometers, Density Meters, Melting Point Devices, Quantos.

Step 4: Software Care — Standard, Comprehensive, or On-Demand service.

License Summary & Activation

- Each system needs an Edition license key + one Instrument license per instrument.
- Non-activated LabX runs 45 days (Demo); cannot use after 60 days.
- Online Activation: requires internet — use <http://activation.mt.com>.
- Offline Activation: generate License Key File → transfer to internet PC → upload to Portal.
- Reactivation required within 10 days if licenses added/removed or hardware replaced.

4. USER MANAGEMENT & ROLE CONFIGURATION

Overview

Define User Roles and Permissions to control access. Located at System workspace > User Management. Supports Local users and Active Directory (AD)-linked users.

Managing Users

- Add/change/delete/temporarily disable users.
- Default local user at installation: PredefinedUser.
- Add Local User: System > User Management > User > New User > fill details > save.
- AD Users: Delegates user/password management to AD. With Role Mapping enabled, roles auto-synced on first login.

Roles

- Roles define rights available to users — each user needs at least one role; multiple roles are cumulative.
- Default: 5 roles — can be added, edited, deleted.
- Rights subdivided by system and modules.

5. SIGNATURE & RELEASE POLICIES

- Global settings — apply to ALL LabX users on all workstations.
- Objects cannot be released until all required signatures obtained.
- Types: None (immediate Approved), Review (1 signature), Review–Approve (2 signatures).
- Auto Release: when enabled, versions release automatically with last required signature.

6. WORKSPACES

Workspace	Purpose	Key Functionalities
Data and Results	Organizing experimental data; storing & viewing results	Data Visualization, Organization, Result Management, Data Integrity & Security
Analysis and Method	Conducting data analysis; developing analytical methods	Analysis Tools, Method Development, Workflow Mgmt, Compliance
Resources and Instruments	Instrument Management & Resource allocation	Instrument Connection/Control, Resource Mgmt, Scheduling
System	Audit Trail, Background Jobs, User Management	Audit Trail, Background Jobs, User Mgmt/Roles, Task Import/Export

7–8. INSTRUMENTS & COMMUNICATION SETUP

Supported METTLER TOLEDO Instruments

Balance, Density Refractometer, Seven Excellence, Thermal Value, Titrator, UV/Vis, Quantos, Peripheral Devices (Barcode reader, Fingerprint reader).

Initial Setup on Instrument

- Language Settings: Select desired language from instrument panel.
- Date/Time Setup: Choose display format; additional steps vary by instrument type.
- Network Settings: Set IP, Subnet, Gateway if using Ethernet; enable DHCP reservation if needed.

- LabX Connection: Define communication type, connection type (network/USB), port (default 8014).
- After restart, instrument panel shows 'Connecting to LabX...'

Note: Date/time on instruments synchronized with LabX Instrument Server at every connection. UI language changed to LabX installation language on connection.

9. ADDING AN INSTRUMENT

Login to LabX Client with 'Edit Resources' rights → Resources > Instruments → New Instrument. Wizard guides: select Instrument Category → Connection Type (Ethernet/Serial/USB) → Instrument Server → enter settings → LabX tests connection → enter name/alias → Finish.

Note: LabX is licensed per connected instrument — one license per instrument only. At least one free instrument license must be available.

10. MANAGING INSTRUMENTS

State	Meaning
Ready	Available, license installed, compatible, no hardware differences
Busy	Instrument is busy
Interaction Required	User interaction needed before task can continue
Blocked	Instrument is blocked
Connected	Instrument is connected to LabX
Error / System error	Error occurred on instrument
Connecting	Currently connecting to LabX
Disconnected	Instrument cannot be reached

Network Interruption: Instrument disconnects; active task paused. Data from completed steps stored in DB. On reconnection, user can resume or restart. No data loss.

11. INSTALLATION

- LabX delivered on write-protected USB flash drive.
- Standalone: All components on one computer.
- Distributed: Application components across different computers.
- Run Setup file (LabXSetup.exe) → Install .NET Framework 4.8 → reboot → select LabX System type.
- Standalone: SQL Server installed automatically. Distributed: Select Custom path → modify parameters.

12–13. DATA MANAGEMENT, ANALYSIS & WORKFLOW

Data Management

- Automatic raw data recording from instruments in proprietary format.
- Central DB with user access control, audit trails, encryption.
- Import/Export in various formats; versioning & backup; metadata management.

Data Analysis & Workflow

- Built-in: statistical calculations, curve fitting, trend analysis, data visualization.
- Customizable & exportable reports; user-defined calculations; scripting & automation.
- Compliance: 21 CFR Part 11, GLP, GMP via compliance templates and electronic signatures.
- Integration with LIMS: seamless data transfer for sample management and traceability.

14–15. DATA BACKUP & MIGRATION

Backup Options

- Automatic Scheduled: regular backups at desired intervals to specific location.
- Automatic Real-time Replication: continuous mirror copy on secondary server.
- Manual Export: entire LabX DB (.mdf/.bak) or specific projects.
- Best Practices: periodic test restores, monitor logs, use encryption, maintain Disaster Recovery Plan.

Migration

- Upgrading LabX Version / Moving to new system / Relocating DB to new server.
- Use METTLER TOLEDO Migration Tool or LabX DB Maintenance Tool (Restore DB with .bak file).
- Example v9→v11: backup .bak → transfer to new DB Server → use 'Publish Archive' → view archived data in 'View Only' mode.

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— CDS (Chromatography Data System) Reference —

16. CDS — BASICS & FUNDAMENTALS

What is CDS?

A computer-based system to control chromatographic instruments, acquire detector signals, process data, and generate results and reports. Ensures data integrity and regulatory compliance (21 CFR Part 11). Controls GC, LC (HPLC), CE, IC, GPC/SEC, GC-MS, LC-MS systems.

Evolution of CDS

Era	Characteristics
Pre-1970s — Manual Era	Paper strip-chart recorders; peaks measured manually using rulers and planimeters
1970s–1980s — Electronic Integrators	Stand-alone integrators for automated peak area and height calculations
1980s–1990s — PC-Based CDS	Computers connected to instruments for digital data acquisition, integration, and storage
Late 1990s–2000s — Client-Server CDS	Networked systems with centralized servers, user access control, audit trails, 21 CFR Part 11 compliance
2010s–Present — Enterprise & Cloud CDS	Centralized, scalable CDS integrated with LIMS/ELN; high data integrity, compliance, remote access, smart analytics

Basic Requirements of a CDS

- Manage growing volume of chromatography data quickly and efficiently.
- Expandable architecture: stand-alone, thin-client, or client/server environments.
- Maintain data adherence to GxP / 21 CFR Part 11.
- Acquire, process, report, review, and approve data in a streamlined series of operations.
- Historical data readily available at any time during retention period.
- All data modifications tracked efficiently.

Globally Adopted CDS Solutions

CDS Solution	Approx. Market Share
Empower™ (Waters)	~35%
Chromeleon™ (Thermo)	~25%
OpenLab™ (Agilent)	~16%
LabSolutions™ (Shimadzu)	~13%
Others (Compass, Clarity, TRILUTION...)	~11%

Note: Worldwide ~70–80% usage share: Empower, Chromeleon, OpenLab, LabSolutions.

17. INTRODUCTION TO CHROMATOGRAPHY

Term coined by Russian botanist Mikhail Tswett (1906). Chromatography separates mixture components by differences in affinity between a mobile phase and an immiscible stationary phase. Used in pharmaceutical, petroleum, food & beverage, cosmetic, forensic, and diagnostic industries.

Chromatographic Techniques

LC/HPLC, Gas Chromatography (GC), Ion Chromatography, Capillary Electrophoresis (CE), Thin Layer Chromatography (TLC), Paper Chromatography, Gel Permeation/Size Exclusion Chromatography (GPC/SEC).

Detectors

LC: UV-Vis, Photo Diode Array (PDA), Refractive Index* (RI), Fluorescence, Charged Aerosol (CAD), Conductivity, Mass Spectroscopy*.

GC: Flame Ionization (FID), Electron Capture (ECD), Thermal Conductivity* (TCD), Flame Photometric (FPD), Mass Spectroscopy*.
(* = Universal Detectors)

18. BASICS OF HPLC & GC INSTRUMENTS

HPLC Components

Solvent Reservoir → Degasser → Pump → Injector → Column (in Column Oven) → Detector → Recorder/Data System → Waste.

Typical Systems: Waters Alliance, Agilent HPLC/UPLC, Shimadzu HPLC, Waters UPLC.

GC Modules

- Carrier gas and other gases (per detector type).
- Samplers: Injector and Headspace.
- Inlet(s), Column and Column Oven.
- Detector (FID, TCD, ECD, etc.).
- Data Acquisition & Control System (e.g., OpenLab CDS, Empower CDS, Chromeleon CDS).

Typical GC Systems: Agilent 7693 Series ALS, Perkin Elmer, 7697A Headspace Sampler.

19. AGILENT OPENLAB CDS — OVERVIEW

Agilent OpenLab CDS is an integrated software platform designed to acquire, control, process, and manage chromatographic data across laboratories (~16% global market share).

Key Highlights

- Centralized data system for LC, GC, LC/MS, and GC/MS.
- Supports Agilent and multi-vendor instruments — Thermo, Shimadzu, Perkin, Metrohm, Waters.
- Combines instrument control, data processing, and reporting.
- Designed for regulated and non-regulated environments.
- Enables secure, compliant, and efficient laboratory operations.
- Compliant with 21 CFR Part 11, GxP, GLP regulatory requirements.

Key Features & Business Value

Category	Details
Core Features	Centralized data management with secure access control; Regulatory compliance: 21 CFR Part 11, audit trails, e-signatures; Scalable: Standalone, Client/Server, or Cloud; Intuitive UI; Integration with OpenLab ECM, LIMS, ELN
Business Value	Improved data integrity and compliance; Increased lab productivity and standardization; Reduced IT complexity and operational risk; Supports digital transformation of QC and R&D; labs

*Note: *21 CFR Part 11 refers to Title 21 of the U.S. Code of Federal Regulations, Part 11 — Electronic Records & Signatures.*

OpenLab CDS Product Comparison

Feature	OpenLab Client/Server	OpenLab CDS (2.x)	OpenLab ChemStation
Category	IT Architecture / Topology	Software Application (New)	Software Application (Legacy)
Primary Goal	Centralization & Networking	Data Integrity & Ease of Use	Flexibility & Familiarity
User Interface	N/A (Depends on App)	Modern Ribbon-style	Classic Windows-style
Instrument Control	Managed by AICs	Integrated (LC/GC/MS)	Integrated (Highly Flexible)
Latest Version	v2.8 (Server/ECM XT)	v2.8	LTS 01.11
OS Support	Windows Server 2022 / Cloud	Windows 10/11 / AWS / Azure	Windows 11 (LTS only)

Feature	OpenLab Client/Server	OpenLab CDS (2.x)	OpenLab ChemStation
Compliance	Highest (Centralized)	Built-in by design	Add-on (requires OpenLab ECM)

Business Integration Scenario

OpenLab CDS bridges LIMS and Chromatograph instruments: LIMS provides sample information → CDS handles instrument control, sample set, data acquisition, data processing, and data storage → Chromatograph instruments execute runs.

Note: Goal of CDS: assure regulatory compliance, protect intellectual property, minimize risk of data loss, support costs, and maximize business values.

20. OPENLAB ECM — SDMS OVERVIEW

Laboratory instrumentation produces large amounts of data (machine and human readable) used throughout drug discovery and development. This data represents significant intellectual property. OpenLab ECM (from Agilent Technologies) is a Knowledge Engineering System designed to store, organize, and protect critical data — including binary data files and human readable instrument output.

Why SDMS is Needed — Data must be:

- Collected, warehoused, and maintained in a central storage location accessible to scientists for review.
- Secure and handled in a manner that meets regulatory and patent guidelines.
- Traceable from upload to archive through a single source.
- Accessible for restoration so review and analysis can be performed within the original environment.

Data Transfer Methods to OpenLab ECM

- Electronic data files can be "printed" to an electronic file and transferred via ECM Print Services.
- Files can be manually uploaded/downloaded by the user.
- Files can be uploaded automatically by an OpenLab Scheduler Agent.

21. OPENLAB SYSTEM ARCHITECTURE

OpenLab ECM has an n-tier architecture allowing modular functionality. The topology consists of an Oracle database server, an Application Server, a Web server, File Transfer servers, a file storage device, and an archived file storage device. It can also include one or more Scheduler Agent devices.

Architecture Components

Component	Function
Web Server	Hosts the application component; configured under IIS using http/https based on user requirement.
File Transfer Server	Used to transfer files to primary storage.
Application Server	Used to configure application components.
Scheduler Agent Server	Used to set up tasks for file sweep.
File Storage	Primary storage location to keep physical files on a permanent basis.
Database	Used to keep application information in database.

22. OPENLAB ADMINISTRATION MODULE

Account Administration

- Configuration — system-wide settings.
- Users/Group — manage user accounts and groups.
- Storage — manage storage configurations.
- Electronic Signature — configure e-signature settings.
- Keys — manage encryption/access keys.
- User Preferences — individual user settings.

Global Administration

- Configure Email — set up email notifications.
- Create Account — create new system accounts.

- License Manager — manage product licenses.
- System Errors — view and manage system error logs.
- System Status — monitor overall system health.

23. USER ACCESS MANAGEMENT (OpenLab ECM)

Three-tier model for access control:

Entity	Purpose
Users	Used to create individual user accounts with personal credentials.
Groups	Used to create user groups to organize users with common roles/permissions.
Roles	Used to define roles and associate them with user groups to grant specific access rights.

Default Roles in OpenLab ECM

- Approver
- Archive Administrator
- Archive Agent
- BPMS User
- Contributor
- Folder Administrator
- Reader
- Retention Manager
- Scheduler Administrator
- System Administrator
- ESignatures

Note: Roles can be reset to defaults using the 'Reset Roles' button in the Roles administration panel.

24. ECM CLIENT INSTALLATION

OpenLab ECM client works with Google Chrome and IE browser. However, Agilent recommends ECM Classic client features to access the application from a local system. End Users need to install ECM Client on the local system in order to access the classic client.

Installer Files to be installed on user's machine:

- AGTBHELRL Reranme.msi
- AGT-HA (File).msi
- AGT301 Client Chrome Installation.msi
- AGTWS-V1 Client-Client Installation Update Tool.msi
- AGTWS-VCI Client Installation Update Server.msi
- AGT4U-TITU Client Uninstallation Update.msi
- AGT v2.01 Application Setup Installer.msp
- FBAETCV Installation Agilent Installation.msi
- AGTMS Administration (Application).msi
- WhatsIm (ie (web)).exe
- LL-MMI Preancitrate.txt (XML Document)

Note: Exact file names/versions may vary by deployment. Consult your system administrator for the correct installer package.

25. COA, ELECTRONIC RECORDS & SIGNATURES

Certificate of Analysis (COA)

For every sample, a Certificate of Analysis (COA) is required before any product can be released to the open market for sale or usage. It lists the parameters, details of tests performed, and methods used. A COA verifies that the product has been tested against defined parameters and complies with regulatory and quality standards.

21 CFR Part 11 — Electronic Records & Signatures

Requirement	Rule
Electronic Records	Must be accurate, complete, protected, and retrievable at all times.
Electronic Signatures	Legally binding if they meet FDA requirements under 21 CFR Part 11.

ECM vs CDS — Core Functional Difference

System	Primary Role
OpenLab ECM (SDMS)	Data archival platform — ensures secure long-term storage and compliance with regulatory requirements.
OpenLab CDS	Data acquisition platform — captures and manages chromatographic data for analysis and reporting.

26. CDS COMPARISON: EMPOWER vs CHROMELEON vs OPENLAB

Feature / Criteria	Waters Empower	Thermo Chromeleon	Agilent OpenLab CDS
Vendor Ecosystem Strength	Very strong with Waters instruments; limited openness	Strong across Thermo + good multi-vendor	Strongest openness for multi-vendor (Agilent + 3rd party)
Multi-Vendor Support	Moderate	Good	Excellent
Architecture	Traditional client-server; database-centric (Oracle)	Client-server + domain-based architecture	Client-server + evolving toward cloud (OpenLab XT)
Cloud Readiness	Limited (mostly on-prem)	Limited / partial	Strong roadmap (OpenLab XT cloud-native)
Scalability (Enterprise)	Proven for large labs but complex to scale	Good scalability	Very strong, especially with XT backend
Ease of Deployment	Moderate (DB-heavy setup)	Easier than Empower	Moderate but improving with standardization
Data Management	Central DB; tight but rigid	Flexible data vaults	Highly flexible + modern data handling
User Interface	Older UI, less intuitive	Modern, intuitive UI	Modern UI (especially newer versions)
Method Development	Strong chromatographic capabilities	Excellent (especially IC workflows)	Strong, especially in HPLC/GC workflows
Instrument Control	Best for Waters instruments	Excellent for Thermo	Broadest support including legacy systems
Compliance (21 CFR Part 11)	Industry gold standard	Strong	Strong
Audit Trails & Data Integrity	Very robust, mature	Robust	Robust, more flexible
Report Customization	Good but sometimes rigid	Very flexible	Very flexible
Integration (LIMS/ELN)	Possible but can be complex	Good integration options	Strong enterprise integration (APIs, connectors)
Customization Capability	Limited	Moderate	High
Performance (Large Data)	Can slow due to DB dependency	Good performance	Strong, especially with XT architecture
Remote Access / Web	Limited	Partial	Improving; stronger in XT/cloud model
Upgrade Complexity	Complex (DB upgrades painful)	Easier than Empower	Moderate
Total Cost of Ownership	High (infra + DB + support)	Moderate	Moderate to high (better ROI in large setups)
Best Fit Use Case	Regulated labs (QC, pharma)	Labs using Thermo ecosystem	Large enterprises, multi-site, cloud adoption

27. CDS PLATFORM — STRENGTHS & LIMITATIONS

Platform	Best For	Limitations
Waters Empower	Highly regulated QC labs; organizations heavily invested in Waters; proven compliance with minimal audit risk.	Not cloud-friendly; less flexible for multi-vendor environments; higher maintenance overhead.
Thermo Chromeleon	Balanced usability + compliance; labs with Ion Chromatography (IC) focus; mixed vendor setups at moderate scale.	Not as future-ready for cloud as OpenLab; slightly less enterprise modularity.
Agilent OpenLab CDS	Enterprise-scale multi-instrument environments; digital transformation / cloud roadmap; integration-heavy ecosystems (LIMS, MES, ELN).	Initial implementation complexity; requires architecture planning (especially XT/cloud).

28. CDS DECISION MATRIX

Scenario / Environment	Recommended CDS
Strict regulatory QC lab, stable on-prem infrastructure	Waters Empower
Balanced usability + compliance, moderate scale, Thermo instruments	Thermo Chromeleon
Large enterprise, cloud roadmap, multi-vendor, integration-heavy	Agilent OpenLab CDS
Ion Chromatography (IC) focused workflows	Thermo Chromeleon
Multi-site deployment with centralized data governance	Agilent OpenLab CDS (XT)

Summary: Empower = stability & compliance (traditional stronghold). Chromeleon = usability + balanced capabilities. OpenLab = future-ready, scalable, enterprise platform.

29. OPENLAB CDS 2.x — CORE CONCEPT

OpenLab CDS 2.x is a modern chromatography data system built around a central workstation that connects to instruments (GC, LC) and provides five integrated capabilities:

Capability	Description
Control	Direct instrument control from the CDS workstation.
Data	Acquires raw signal data from connected instruments.
Report	Generates analytical reports from processed results.
User Management	Centralized user accounts, roles, and access control.
Audit Trail	Tracks all actions, modifications, and approvals for compliance.
Single Sample & Sequencing	Supports both single sample runs and full sequence batches via the Acquisition client.

30. OPENLAB CDS — BASIC WORKFLOW

The workflow splits into two client roles:

Client	Role	Key Functions
ACQ (Acquisition) Client	Used during data collection	Set ACQ Method; run Single Sample or Sequences; instrument control.
DA (Data Analysis) Client	Used after data collection	Integration of peaks; Calibration; Reporting of results.

Workflow sequence: Sample → Instrument → Signal → ACQ Method (acquisition) → .acam file → Processing Method → Integration → Result → Report.

31. OPENLAB CDS TOPOLOGY

OpenLab CDS v2.x provides 3 main topology types, with 4 sub-types under Client/Server:

Topology	Storage	Instruments	Description
Workstation	Local Windows file system	Up to 4	All components (Data Analysis, Acquisition, Instrument Control, OpenLab Platform) on one PC.
Workstation Plus	Content Management Database	Up to 4	Same as Workstation but storage replaced by Content Management database for better data control.
Client/Server — Basic (All-in-one)	Content Management	Up to 6 per AIC	All core services (DB, repository, file storage) on a single server. AICs collect data; clients connect for analysis. Suited for small to medium labs.
Client/Server — 2-Server	Content Management	Scalable	Application server with external DB server and local file storage, OR application server with local DB and external file storage.
Client/Server — 3-Server	External DB + External File	Scalable	Application server (OpenLab/ECM XT, Shared Services, Secure Storage) + separate Database server + separate File Storage server.
Client/Server — Enterprise	External DB + External File	Large scale	As 3-server plus dedicated Search Server for indexing and Load Balancer to distribute traffic among multiple application servers.

Note: AIC = Agilent Instrument Controller. Each AIC connects up to 6 instrument points (standalone/virtualized PC) or 4 points (networked workstation).

32. OPENLAB CDS — INSTALLATION WORKFLOW

The installation follows a 5-step sequential workflow:

Step	Task	Key Actions
1	Prepare PC & Instruments	Verify OS & hardware; configure instruments; set up network & admin readiness.
2	Install OpenLab CDS Software	Install core application; install drivers & add-ons; run verification tools.
3	Post-Installation Tasks	Configure user accounts; verify services are running; set firewall & AV exceptions.
4	License Activation	Subscribe/Agilent login; apply licenses; validate activation status.
5	Configure Workstation	Apply security policies; configure users & roles; add projects & instruments.

33. PC CONFIGURATION & SYSTEM PREPARATION TOOL

Recommended Hardware (OpenLab CDS Workstations)

Item	CDS Workstation	CDS Workstation Plus
Tested Agilent PC	HP Z2G9 SFF; Intel Core i5-12500, 3.0GHz, 18MB cache, 6 Core	HP Z2G9 SFF; Intel Core i5-12500, 3.0GHz, 18MB cache, 6 Core
CPU (minimum)	Intel i5, i7, or Xeon E3; 3.0 GHz+, 4 Core	Intel i5, i7, or Xeon E3; 3.0 GHz+, 4 Core
RAM	1–2 instruments: 8 GB; 3+ instruments: 16 GB (4 GB reserved for OS)	16 GB DDR5-4800 MHz RAM (2x 8 GB)
Hard Disk	1x 500 GB 7200 RPM SATA (min) or equivalent SSD (recommended)	1 TB 7200 RPM SATA (min); if disc array: 2x 1TB in RAID1 + SSD
Graphics	1600×900 minimum; 1920×1080 recommended	Same
RS-232 Port	1 serial port for instruments using RS-232 communication	Same
USB Port	USB 2 required for installation via provided media	Same
LAN Card	100 MB/1 GB LAN; 2nd LAN card recommended to isolate instrument data traffic	Same

Supported Operating Systems

Windows Product	Workstation	Networked Workstation	Client	AIC
Windows 11 64-bit, Professional or Enterprise	✓	✓	✓	✓
Windows 10 64-bit, Professional or Enterprise	✓	✓	✓	✓
Windows Server 2016 64-bit Standard or Datacenter	✗	✗	✗	✗
Windows Server 2019 64-bit Standard or Datacenter	✗	✓ ¹	✓ ²	✓ ¹
Windows Server 2022 64-bit Standard or Datacenter	✗	✓ ¹	✓ ²	✓ ¹

Note: ¹ Supported, but not recommended. ² Virtual environment.

System Preparation Tool (SPT)

- To open the installer: right-click setup.exe and run as Administrator.
- Run System Preparation Tool (SPT) from the Planning section before installation to shorten install process and avoid PC reboots.
- SPT inspects and applies recommended Windows system settings automatically via PowerShell scripts.
- Select the product configuration as the first step (e.g., OpenLab CDS 2.x, Server Win11).
- SPT conducts PASS/FAIL checks: PASS = setting already applied; FAIL = needs to be addressed.
- System Preparation Report summarizes OS status — mandatory settings in 'Action Required' section must be resolved before installation.
- PASS = tool changed the setting automatically; FAIL = tool could not change the setting (manual action required).

34. OPENLAB CDS — INSTALLATION DETAIL

Installation Steps

- Installer offers three modes: Installation, Configuration, Uninstall.
- Accept license agreement ('I agree with the terms and conditions') and click Next.
- Installation Folder: contains the Agilent OpenLab CDS application and all subcomponents — leave default paths unless customer requests different location.
- Storage Configuration: default Projects Root Path is C:\CDSPROJECTS — verify with customer; location must have adequate storage.
- Data Repository: internal storage infrastructure for diagnostic and topology data. Set a password — required for re-installation using existing repository database.
- Installer automatically applies mandatory Windows settings for proper installation.

- Warning displayed if manual OS actions are required — options: Cancel Installation (abort & fix), Resume Installation (continue despite pending settings), Open SPT Report (review required actions).
- After installation completes, click Next. Review System Preparation report link — if manual settings still needed, a warning is displayed.

Post-Installation Tasks

- Click 'Run Software Verification' to confirm everything installed correctly.
- Software Verification Report shows: date, process name, run type, product name, summary (pass/fail), and detail summary.
- Check Programs and Features to confirm all Agilent OpenLab CDS components are listed (e.g., Agilent OpenLab CDS, Data Player, Instrument Tool, LC/MS, GC/MS, Certificate Service, CopyTo Client, Data Collection Agent, DataReporty, Distributed Transaction Coordinator, Proxy Config Service, Revenue Proxy, Shared Services 64-bit, Storage Client Services, Software Verification Tool, Tool Service, Test Services CDS Plugin).

35. OPENLAB CDS — LICENSING MODEL

Key Licensing Terms

Term	Definition
Instrument	An analytical system made up of one or more modules configured together to achieve desired analytical functionality (e.g., LC/MSD system).
Points	The relative level of load (CPU and RAM consumption) an instrument or module puts on a physical/virtual system. Different detector types carry higher point values.
Concurrent Sessions	Number of simultaneous connections to the system — includes user and instrument connections. Use points as the measure (reflects both number of instruments and data generated per instrument type).
Instrument Licenses	Connection licenses required in the software to run a specific instrument configuration. Note: Points do NOT equal instrument connection licenses.

Points-Based Capacity — Workstations

OpenLab CDS Configuration	Supported Instrument Points
Workstation (with local file storage)	Up to 4 points
Workstation Plus (with local secure storage)	Up to 4 points
VL Workstation (with local file storage)	1 instrument; up to 2 points
VL Workstation Plus (with local secure storage)	1 instrument; up to 2 points

Points-Based Capacity — AICs

OpenLab CDS AIC	Instrument Points per AIC
Standalone PC	Up to 6 points
Virtualized PC	Up to 6 points
Networked Workstation	Up to 4 points
Client/Server	6 points per AIC

Points & Connections Examples

System	Points (Capacity)	Connections (Licenses)
LC + DAD	2 points	2 Connections (1x LC + 1x LC 3D)
LC + HDR DAD	3 points	2 Connections (1x LC + 1x LC 3D)
LC + DAD + FLD	2 points	2 Connections (1x LC + 1x LC 3D)
LC + MS	3 points	2 Connections (1x LC + 1x MS)
LC + MS + DAD	4 points	3 Connections (1x LC + 1x MS + 1x LC 3D)
GC with HS	1 point	1 Connection (1x GC; HSS is 0 connections)

System	Points (Capacity)	Connections (Licenses)
GC + MS	3 points	2 Connections (1x GC + 1x MS)

Note: Agilent OpenLab CDS 2.x uses a centralized feature-based licensing model. Licenses are consumed only by instruments, detectors, and licensed options. Servers and users do NOT consume licenses — so scaling application, database, or search servers has no licensing impact.

License Registration Steps

Step	Action
1. Start SubscribeNet	Go to https://Agilent.SubscribeNet.com
2. Registration Page	Click to navigate to the registration page.
3. Register Authorization Code	Use one Authorization Code; enter all customer information. The email address becomes your login ID.
4. Receive Email	Welcome email with password sent to the provided email address.
5. Login to Customer Account	Login using email address and password received.
6. Register Authorization Code	Enter new authorization code and make available the new license entitlements.
7. Generate or View License	Follow prompts to generate license. You will be prompted for the HOST NAME of the computer (no DNS suffix).
8. Download License	Download license file and save it.
9. Install License	Add license via Control Panel → Add Licenses → select license file or paste contents → OK.

36. CREATE & CONFIGURE INSTRUMENT IN OPENLAB CDS

Steps to Create an Instrument

Step	Action
1	Click the Control Panel icon to start the Control Panel.
2	Click on Instruments in the left navigation panel.
3	Click the Create icon in the toolbar.
4	Select 'Create Instrument' from the dropdown.
5	Enter instrument Name, IP Address, Description, Application (e.g., OpenLab CDS), Instrument Controller (e.g., Mx767WS), Instrument Type (e.g., Agilent GC & GC/MS Systems), Contact, and Default Project.
6	Click OK to save the instrument.

Steps to Configure an Instrument

Step	Action
1	Click the newly created instrument in the Instruments tree.
2	Click 'Configure Instrument' in the toolbar.
3	In the Agilent GC & GC/MS Systems dialog, select the proper instrument module from the Available Modules list.
4	Double-click the selected instrument module (or click the arrow) to move it to Configured Modules.
5	Right-click the configured module and select Open (or double-click) to open the configuration dialog.
6	Enter the GC's IP Address in the Connection Info panel and click OK.

37. OPENLAB CDS — USER MANAGEMENT & ROLES

User Creation Modes

Mode	Description
Internal	The only mode where new users can be created directly within the system. In all other modes, users must already exist in a different system and are mapped/imported.

Mode	Description
Windows Domain	Imports existing Windows users into OpenLab CDS. Authentication is handled by Windows Active Directory domain.

Role Types in OpenLab CDS

Role Type	Applied To	Examples
Project Roles	Projects and Projects group in OpenLab Control Panel	Project Administrator role, Contributor role (Workstation Plus)
Instrument Roles	Instruments and Instrument Locations in OpenLab Control Panel	Instrument User role, Chemist or Technician roles
Administrative Roles	OpenLab Control Panel and Content Management (Workstation Plus)	Archivist role

Key Role Privileges

- Project Management — Project Roles & Group management.
- E-Signature — access to e-signature/approval functions.
- Sample Prep — access to sample preparation-related functions.
- Acquisition Method — access to method editing functions.
- Processing Method — access to processing method-related functions.
- Report Template / Sequence Template — access to report/sequence designer functions.
- Audit Trail — access to audit-trail-related functions.
- Control — access to instrument control-related functions.
- Data Processing — access to data processing-related functions.
- File/Folder Operations — access to file and folder operations.
- Peak — access to peak/result functions.
- Custom Tools — access to custom tools.
- Snapshot — access to snapshot functions.